What is kubelet?

Professional Answer

kubelet is an agent running on each worker node that ensures containers are running as defined in Pod specifications.

Unprofessional Answer

kubelet runs pods.

What is kubelet and what are its responsibilities?

kubelet is an agent that runs on every worker node in the cluster.
It receives pod specifications from the API server.
kubelet ensures containers are created and running as defined in the pod.
It monitors container health using probes and restarts containers if needed.
kubelet also reports node and pod status back to the control plane.

How does kubelet communicate?

Professional Answer

kubelet communicates with the API Server to receive Pod specifications and reports node and pod status back.

Unprofessional Answer

kubelet talks to Kubernetes.

kubelet (Node Agent Internals)

What kubelet is internally

- Long-running daemon
- Runs on every worker node

Internal Responsibilities

1. **Pod Lifecycle Management**
    - Reads PodSpec
    - Creates containers
    - Starts/stops containers

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2. **Container Health**
    - Executes liveness/readiness/startup probes
3. **Node Status Reporting**
    - Sends heartbeats to API server
    - Reports resource usage
4. **Volume Management**
    - Mounts/unmounts volumes

kubelet does NOT

Schedule pods
Expose services

Perfect 👍
kubelet is *even more important than kube-proxy* in interviews because it touches **containers, Linux, networking, storage, health checks, and failures**.

Below is a **complete senior-level kubelet package**:

- Interview Q&A (basic → advanced)
- Troubleshooting (step-by-step, command driven)
- Real production scenarios
- Ultra-hard interview questions
- Killer one-liners

Kubernetes kubelet – Interview Q&A + Troubleshooting + Scenarios

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PART 1️⃣ Core Interview Questions (Must-Know)

Q1. What is kubelet?

✅ **Professional Answer**

kubelet is the primary node agent that runs on every worker node.
It ensures that containers described in PodSpecs are running and healthy.
kubelet communicates with the API Server to receive Pod definitions.

It interacts with the container runtime using CRI.
kubelet reports node and Pod status back to the control plane.

❌ **Weak Answer**

kubelet runs pods.

Q2. Where does kubelet run and how?

✅ **Professional Answer**

kubelet runs as a system service on every worker node.
It is not deployed as a Pod.
It starts automatically when the node boots.
kubelet manages Pods locally on that node.
If kubelet stops, the node becomes NotReady.

Q3. How does kubelet communicate with Kubernetes?

✅ **Professional Answer**

kubelet communicates with the API Server over HTTPS.
It authenticates using certificates or tokens.
It watches assigned Pods and node configuration.
It sends heartbeats and Pod status updates.
This communication is continuous and bidirectional.

Q4. Does kubelet run containers directly?

✅ **Professional Answer**

No, kubelet does not run containers directly.
It uses the Container Runtime Interface (CRI).
The runtime can be containerd or CRI-O.
kubelet sends instructions to the runtime.
The runtime handles image pulls and container execution.

Q5. What responsibilities does kubelet have?

✅ **Professional Answer**

kubelet ensures Pods are running as defined.
It performs container health checks.

It mounts volumes into Pods.
It enforces CPU and memory limits.
It reports node health and resource usage.

PART 2️⃣ kubelet & Health Checks (Very Important)

Q6. How does kubelet perform health checks?

✅ **Professional Answer**

kubelet executes liveness, readiness, and startup probes.
Liveness probes decide when to restart containers.
Readiness probes control Service traffic routing.
Startup probes delay liveness checks for slow apps.
Incorrect probes cause frequent restarts.

Q7. What happens if a liveness probe fails?

✅ **Professional Answer**

kubelet kills the container.
It restarts the container automatically.
The Pod itself is not recreated.
This can cause crash loops if misconfigured.
Node health is unaffected.

Q8. What happens if a readiness probe fails?

✅ **Professional Answer**

kubelet marks the Pod as NotReady.
Traffic is removed from Services.
The container is not restarted.
The Pod keeps running.
This protects users from unhealthy traffic.

PART 3 Resource Management & Evictions

Q9. How does kubelet manage CPU and memory?

✅ **Professional Answer**

kubelet uses Linux cgroups.
CPU requests affect scheduling decisions.
Memory limits are enforced strictly.
Exceeding memory limits causes OOMKills.
CPU limits cause throttling, not kills.

Q10. What is Pod eviction and who triggers it?

✅ **Professional Answer**

Pod eviction occurs during resource pressure.
kubelet detects memory, disk, or PID pressure.
It evicts Pods based on priority and QoS.
BestEffort Pods are evicted first.
This protects node stability.

PART 4 kubelet Troubleshooting (Step-by-Step)

Scenario 1 Node is NotReady

Steps

kubectl get nodes

1. SSH into node
2. Check kubelet status

systemctl status kubelet

3. Check logs

journalctl -u kubelet

4. Verify API Server connectivity
5. Restart kubelet if needed

Scenario 2 Pods stuck in ContainerCreating

Root Causes

- Container runtime down

- Disk full
- Image pull failure
- Volume mount issues

Commands

systemctl status containerd

df -h

journalctl -u kubelet

Scenario 3 Node Ready but Pods keep failing

Investigation

- Check kubelet logs
- Inspect probe failures
- Check OOMKilled events

kubectl describe pod <pod>

Scenario 4 kubelet Running but Node Not Ready

Likely Causes

- Certificate expired
- API Server unreachable
- Network issue

PART 5 Real Production Scenarios 🔥

Scenario A

Pods restart continuously after deployment

Expected Answer

- Misconfigured liveness probe

- App startup time too long
- CPU throttling
- Memory limit too low

Scenario B

Node shows DiskPressure even with free space

Expected Answer

- Inode exhaustion
- Container image filesystem full
- kubelet eviction thresholds misconfigured

Scenario C

Pods evicted even when node has free memory

Expected Answer

- Node allocatable miscalculation
- Reserved system memory too high
- Memory fragmentation

Scenario D

After upgrade, node joins but no Pods start

Expected Answer

- CRI socket mismatch
- kubelet config mismatch
- Runtime API version mismatch

PART 6 — Ultra-Hard kubelet Interview Questions (Senior / Principal)

Q1. Can Pods run if kubelet is down?

Answer:
Existing containers may continue running, but Pods are unmanaged.
No new Pods start.
Health checks stop.
Node becomes NotReady.

Q2 kubelet vs scheduler – who decides what?

Answer:
Scheduler decides *where* Pods run.
kubelet decides *how* they run.
kubelet never schedules Pods.
It only executes assignments.

Q3 Why is kubelet failure more dangerous than kube-proxy failure?

Answer:
kubelet failure stops Pod lifecycle management.
Containers stop being monitored.
Evictions and health checks stop.
The node becomes unusable.

Q4️⃣ How do you debug kubelet when kubectl is down?

Answer:

- SSH into node
- Check kubelet logs
- Inspect static Pod manifests
- Validate runtime health

PART 7️⃣ Golden Interview Lines 🏆

kubelet is the heart of the worker node — if it stops, the node is alive but useless.

Scheduler decides placement, kubelet enforces reality.

Most kubelet issues are Linux or resource issues, not Kubernetes bugs.

PART 8️⃣ kubelet vs kube-proxy (One-Line Comparison)

Component	Responsibility
kubelet	Pod lifecycle & health
kube-proxy	Service networking

If you want next 🚀

- 🔥 **kubelet vs container runtime deep dive**
- 🧠 **Linux internals behind kubelet (cgroups, namespaces)**
- 🎭 **Live mock interview where YOU answer kubelet questions**

Say the word 💪